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CHEMISTRY
STANDARD LEVEL
PAPER 3

SPECIMEN PAPER

1 hour

Candidate session number

Examination code

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INSTRUCTIONS TO CANDIDATES

- Write your session number in the boxes above.
- Do not open this examination paper until instructed to do so.
- Section A: answer all questions.
- Section B: answer all of the questions from one of the options.
- Write your answers in the boxes provided.
- A calculator is required for this paper.
- A clean copy of the *Chemistry data booklet* is required for this paper.
- The maximum mark for this examination paper is *[35 marks]*.

Option	Questions
Option A — Materials	3 – 6
Option B — Biochemistry	7 – 9
Option C — Energy	10 – 12
Option D — Medicinal chemistry	13 – 15

34 pages
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SECTION A

*Answer **all** questions. Write your answers in the boxes provided.*

1. Compounds used to generate cooling in refrigerators and air-conditioning systems are known as refrigerants. A refrigerant undergoes a reversible change of state involving vaporization and condensation. The search for suitable refrigerants has occupied chemists for approximately 200 years.

Previously, the most popular refrigerants were chlorofluorocarbons (CFCs), but these have been replaced, first by hydrochlorofluorocarbons (HCFCs) and more recently by hydrofluorocarbons (HFCs).

Some data on examples of these three classes of refrigerants are shown below.

Class	**Compound**	**ODP**[1]	**GWP**[2] **over 100 years**	ΔH_{vap}[3] **/ kJ mol^{-1}**	**Atmospheric lifetime / years**
CFC	CCl_3F	1.0	4000	24.8	45
CFC	CCl_2F_2	1.0	8500	20.0	102
HCFC	$CHCl_2CF_3$	0.02	90	26.0	1
HCFC	$CHClF_2$	0.05	1810	20.2	12
HFC	CH_2FCF_3	0	1100	–	–
HFC	CHF_2CF_3	0	3500	30.0	32

[1] ODP: The ozone depletion potential (ODP) is a relative measure of the amount of degradation to the ozone layer caused by the compound. It is compared with the same mass of CCl_3F, which has an ODP of 1.0.

[2] GWP: The global warming potential (GWP) is a relative measure of the total contribution of the compound to global warming over the specified time period. It is compared with the same mass of CO_2, which has a GWP of 1.0.

[3] ΔH_{vap}: Defined as the energy required to change one mole of the compound from a liquid to a gas.

(a) (i) Explain why the values for ODP and GWP have no units. *[1]*

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(This question continues on the following page)

(Question 1 continued)

(ii) By making reference to the chemical formulas and ODP values of the compounds, comment on the hypothesis that chlorine is responsible for ozone depletion. *[1]*

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(b) Use data from the table to interpret the relationship between the atmospheric lifetime of a gas and its GWP. *[2]*

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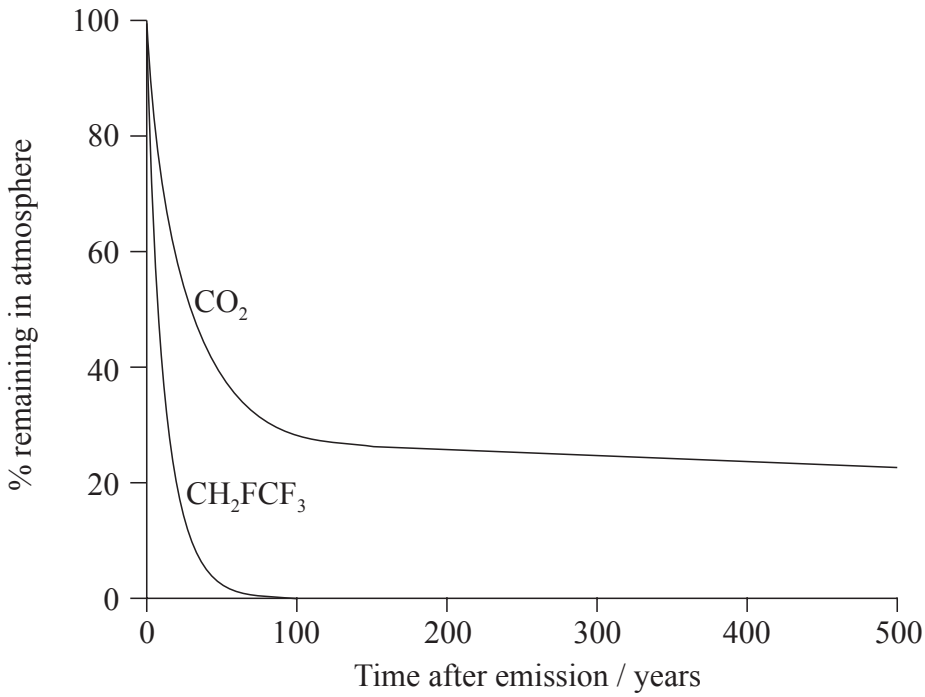
(Question 1 continued)

(c) The graph shows the change in levels with time of equal masses of CO_2 and CH_2FCF_3 introduced into the atmosphere.



(i) Apply IUPAC rules to state the name of CH_2FCF_3. *[1]*

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(ii) The $\Delta H_{vaporization}$ for CH_2FCF_3 is $217\,kJ\,kg^{-1}$. Calculate the value of the enthalpy change for the condensation of one mole of CH_2FCF_3. *[2]*

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(This question continues on the following page)

(Question 1 continued)

(iii) With reference to the graph on page 4, comment on the atmospheric lifetime of CO_2 relative to CH_2FCF_3, and on the likely influence of this on climate change. *[2]*

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2. Thomas wants to determine the empirical formula of red-brown copper oxide. The method he chooses is to convert a known amount of copper(II) sulfate into this oxide. The steps of his procedure are:

- Make 100 cm^3 of a 1 mol dm^{-3} solution using hydrated copper(II) sulfate crystals.
- React a known volume of this solution with alkaline glucose in order to convert it to red-brown copper oxide.
- Separate the precipitated oxide and find its mass.

(a) Thomas calculates that he needs $0.1 \times [1 \times 63.55 + 1 \times 32.07 + 4 \times 16.00] = 15.962 \pm 0.001$ g of the copper(II) sulfate to make the solution. Outline the major error in his calculation. *[1]*

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(b) He now adds $100 \pm 1\,cm^3$ of water from a measuring/graduated cylinder and dissolves the copper(II) sulfate crystals. A friend tells him that for making standard solutions it is better to use a volumetric flask rather than adding water from a measuring cylinder. Suggest **two** reasons why a volumetric flask is better. *[2]*

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(c) Thomas now heats 25 cm^3 of the solution with excess alkaline glucose to convert it to a suspension of red-brown copper oxide. Describe how he can obtain the pure, dry solid product. *[2]*

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(This question continues on the following page)

(Question 2 continued)

(d) Using the same chemical reactions, suggest how Thomas' method to determine the mass of red-brown copper oxide that could be obtained from a known mass of copper(II) sulfate crystals might be simplified to produce more precise results. *[1]*

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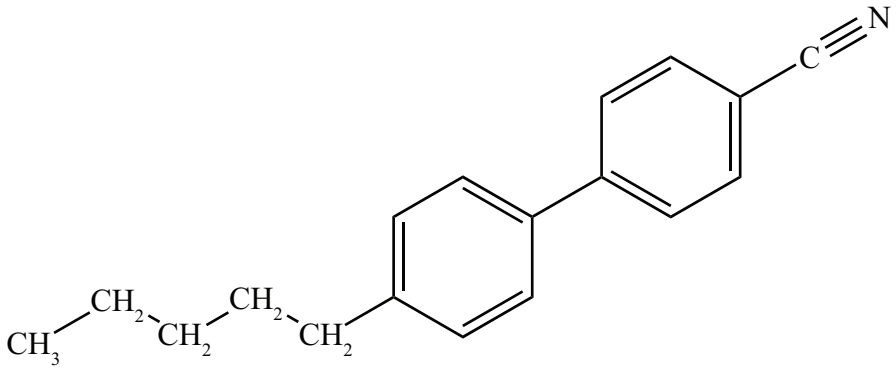
SECTION B

*Answer **all** of the questions from **one** of the options.*

Option A — Materials

3. (a) The molecule shown below is frequently used in liquid-crystal displays (LCDs).



Identify a physical characteristic of this molecule that allows it to exist in a liquid-crystal state. *[1]*

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(b) (i) Describe the chemical vapour deposition (CVD) method for the production of carbon nanotubes. *[2]*

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(ii) Many modern catalysts use carbon nanotubes as a support for the active material. State the major advantage of using carbon nanotubes. *[1]*

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(Option A continues on the following page)

(Option A continued)

4. Different metal oxides are widely used in the production of ceramic materials and their function is closely linked to the type of bonding present in the compound.

(a) Both magnesium oxide and cobalt(II) oxide are incorporated into ceramics. Use section 8 of the data booklet to calculate values to complete the table below. *[2]*

Compound	**Magnesium oxide**	**Cobalt(II) oxide**
Electronegativity difference	. . . . .	. . . . .
Average electronegativity	. . . . .	. . . . .

(b) Predict the bond type and percentage covalent character of each oxide, using section 29 of the data booklet. *[2]*

Compound	**Magnesium oxide**	**Cobalt(II) oxide**
Bond type	. . . . .	. . . . .
% covalent character	. . . . .	. . . . .

(Option A continues on the following page)

(Option A continued)

5. Magnesium is an essential component of chlorophyll and traces of it can be found in various fluids from plants. Its concentration may be estimated using inductively coupled plasma optical emission spectroscopy (ICP-OES).

(a) Outline what the specific plasma state involved in ICP spectroscopy comprises. *[1]*

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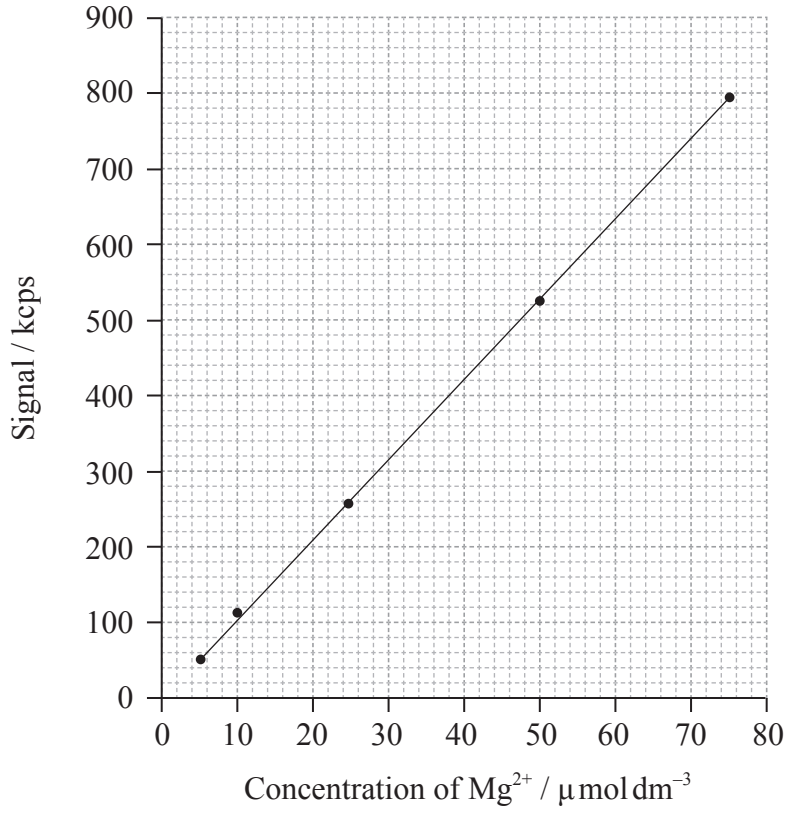
(b) An ICP-OES calibration curve for magnesium is shown in the graph below.



(Option A continues on the following page)

(Option A, question 5 continued)

(i) Determine the mass of magnesium ions present in 250 cm^3 of a solution with a concentration of 10 $\mu mol\ dm^{-3}$. *[2]*

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(ii) Taking into account your answer to part (b)(i), discuss how the solutions for this calibration curve could be produced. *[2]*

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(iii) Two solutions gave count rates of 627 kcps and 12 kcps respectively. Justify which solution could be more satisfactorily analysed using this calibration graph. *[1]*

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(Option A continues on the following page)

(Option A continued)

6. Plastics, such as PVC and melamine, are widely used in modern society.

(a) PVC is thermoplastic, whereas melamine is thermosetting. State one other way in which scientists have tried to classify plastics, and outline why the classification you have chosen is useful. *[2]*

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(b) It was almost a century after the discovery of PVC before Waldo Semon turned it into a useful plastic by adding plasticizers. State and explain the effect plasticizers have on the properties of PVC. *[2]*

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(c) Justify why, in terms of atom economy, the polymerization of PVC could be considered "green chemistry". *[1]*

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(Option A continues on the following page)

(Option A, question 6 continued)

(d) In spite of the conclusion in part (c), many consider that PVC is harmful to the environment. Identify **one** specific toxic chemical released by the combustion of PVC. *[1]*

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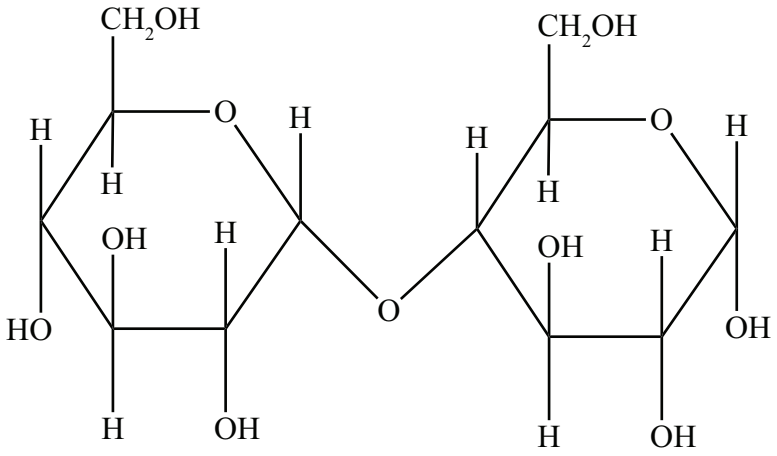
End of Option A

Option B — Biochemistry

7. The diagram below shows the structure of a disaccharide called maltose.



(a) Identify on the diagram one primary alcohol group by marking I on the oxygen, **and** one secondary alcohol group by marking II on the oxygen. *[1]*

(b) (i) Formulate an equation, using molecular formulas, to show the conversion of this molecule into its monomers. *[1]*

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(ii) Identify the type of metabolic process shown in part (b)(i). *[1]*

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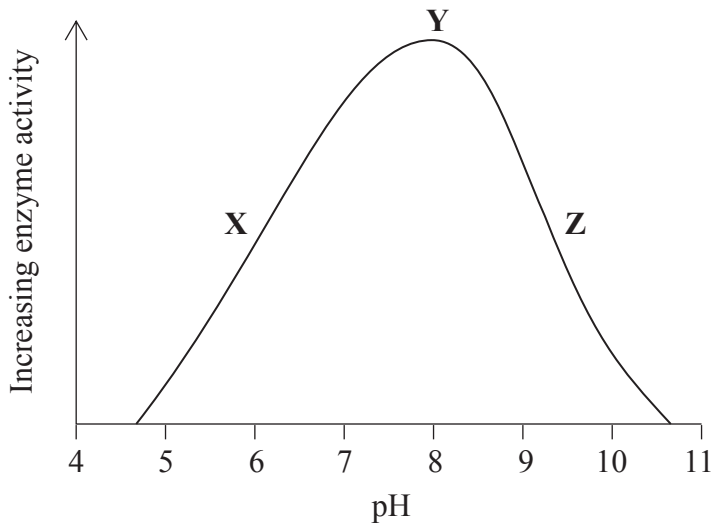
(Option B continues on the following page)

(Option B, question 7 continued)

(c) The reaction in part (b) is catalysed by the enzyme maltase. Experiments were carried out to investigate the rate of breakdown of maltose in the presence of maltase over a range of pH values from 4 to 11. The results are shown below.



Describe how the activity of the enzyme changes with pH, including in your answer specific reference to how the pH is affecting the enzyme at **X**, **Y** and **Z**. *[3]*

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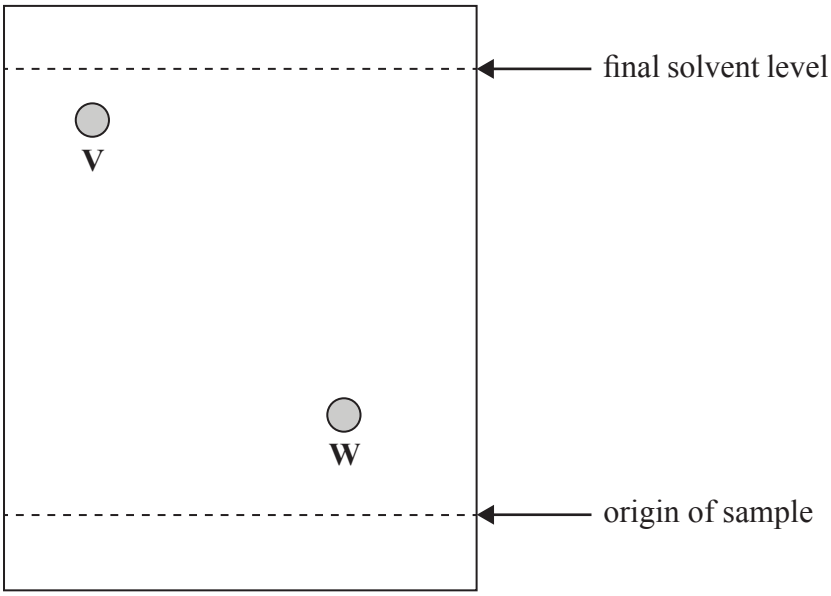
(Option B continues on the following page)

(Option B, question 7 continued)

(d) A separate experiment was done to determine the amino acid composition of maltase. A sample of the enzyme was hydrolysed into a mixture of its component amino acids. Paper chromatography and a locating agent were then used to try to identify the amino acids present in the mixture. The diagram below shows part of the chromatogram in which the positions of two amino acids, **V** and **W**, can be seen.



Amino acid	R_f
Lysine	0.14
Glutamine	0.26
Proline	0.41
Methionine	0.56
Leucine	0.73

Use the chromatogram and the data table to deduce the identity of **V** and **W** if possible. *[2]*

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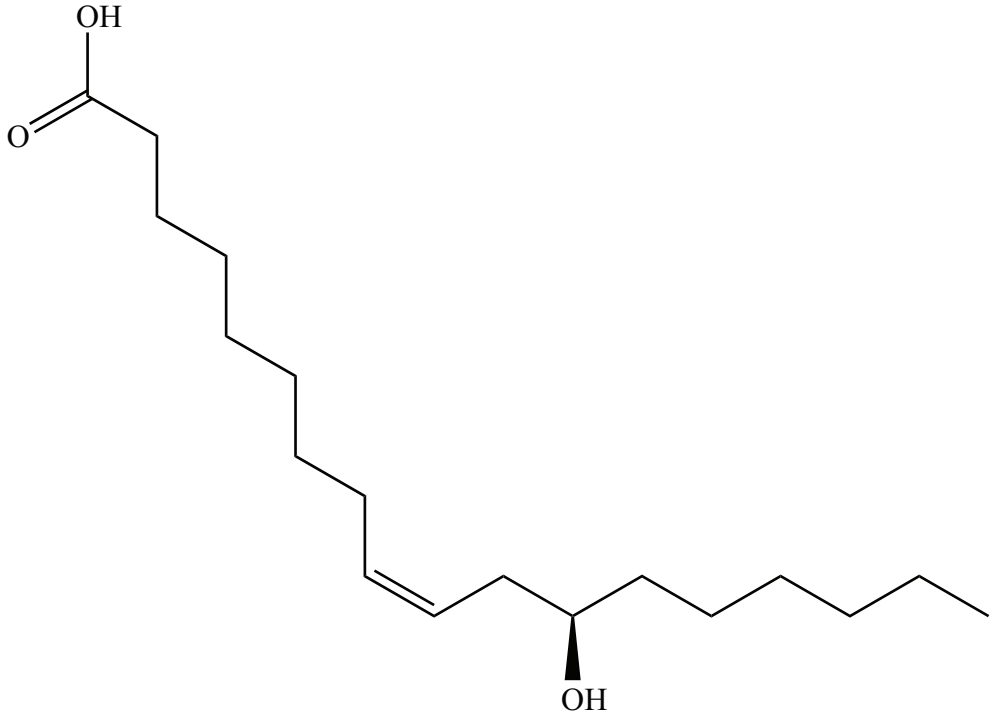
(Option B continues on the following page)

(Option B continued)

8. The castor plant is grown as a crop for its oil. Castor oil is mostly a triglyceride of the relatively rare fatty acid ricinoleic acid, whose structure is given below.



(a) State the molecular formula of ricinoleic acid. *[1]*

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(b) (i) Compare and contrast the structure of ricinoleic acid with stearic acid, whose structure is given in section 34 of the data booklet. *[3]*

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(Option B continues on the following page)

(Option B, question 8 continued)

(ii) State and explain how you would expect ricinoleic acid triglyceride to differ from stearic acid triglyceride in its tendency to undergo oxidative rancidity. *[2]*

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(c) The castor seed contains ricin, a toxic protein which is fatal in small doses. During the oil extraction process, the toxin is inactivated by heating.

(i) Outline why ricin loses its toxic effects on being heated. *[1]*

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(ii) Examine why many countries no longer harvest the castor plant but rely instead on imports of castor oil from other countries. *[2]*

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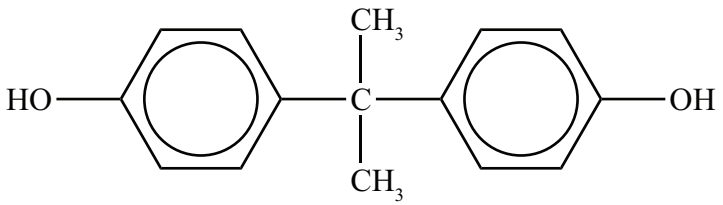
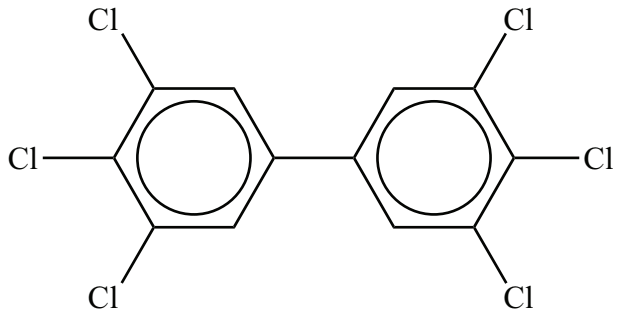
(Option B continues on the following page)

(Option B continued)

9. The figure below shows two examples of molecules known as xenoestrogens, a type of xenobiotic. They have effects on living organisms similar to those of the female hormone estrogen. These compounds are found in the environment and can be taken up by living organisms, where they may be stored in certain tissues.





(a) State what is meant by the term xenobiotic. *[1]*

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(b) With reference to their structures, outline why these xenobiotics are stored easily in animal fat. *[1]*

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(Option B continues on the following page)

(Option B, question 9 continued)

(c) One way to decrease the concentration of a xenobiotic in the environment is to develop a specific molecule, a "host", that can bind to it. The binding between the host and the xenobiotic forms a supramolecule.

State **three** types of association that may occur within the supramolecule between the host and the xenobiotic. *[1]*

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End of Option B

Please **do not** write on this page.

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will not be marked.

Option C — Energy

10. Plants convert solar energy into chemical energy. It would therefore be very convenient to use plant products, such as vegetable oils, directly as fuels for internal combustion engines.

(a) (i) Identify the major problem involved in using vegetable oils directly as a fuel in a conventional internal combustion engine. *[1]*

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(ii) Transesterification of the oil overcomes this problem. State the reagents required for this process. *[1]*

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(b) Plant products can also be converted to ethanol, which can be mixed with alkanes, such as octane, to produce a fuel. The table below gives some properties of these compounds.

Compound	**Molar mass / g mol^{-1}**	**Density / g dm^{-3}**	**ΔH_c / kJ mol^{-1}**	**Equation for combustion**
Ethanol	46.08	789	–1367	$C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$
Octane	114.26	703	–5470	$C_8H_{18}(l) + 12\frac{1}{2}O_2(g) \rightarrow 8CO_2(g) + 9H_2O(l)$

(i) State the name of the process by which ethanol can be produced from sugars. *[1]*

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(Option C continues on the following page)

(Option C, question 10 continued)

(ii) The energy density of ethanol is 23 400 kJ dm^{-3}. Use data from the table to determine the energy density of octane. *[1]*

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(iii) Use these results to outline why octane is the better fuel in vehicles. *[1]*

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(iv) Use data from the table to demonstrate that ethanol and octane give rise to similar carbon footprints. *[1]*

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(v) Outline why, even though they have similar carbon footprints, using ethanol has less impact on levels of atmospheric carbon dioxide. *[1]*

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(Option C continues on the following page)

(Option C continued)

11. Nuclear power is an energy source that does not involve fossil fuels. Current nuclear technology is dependent on fission reactions.

(a) Commercial nuclear power technology developed very rapidly between 1940 and 1970. Outline why this occurred. *[1]*

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(b) The equation for a typical nuclear fission reaction is:

$$^{235}_{92}\mathrm{U} + ^{1}_{0}\mathrm{n} (\rightarrow ^{236}_{92}\mathrm{U}) \rightarrow ^{90}_{38}\mathrm{Sr} + ^{136}_{54}\mathrm{Xe} + 10\,^{1}_{0}\mathrm{n}$$

The masses of the particles involved in this fission reaction are shown below.

Mass of neutron	=	1.00867 amu
Mass of U-235 nucleus	=	234.99333 amu
Mass of Xe-136 nucleus	=	135.90722 amu
Mass of Sr-90 nucleus	=	89.90774 amu

Using these data and information from sections 1 and 2 of the data booklet, determine the energy released when one uranium nucleus undergoes fission. *[3]*

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(Option C continues on the following page)

(Option C, question 11 continued)

(c) The half-lives of components of spent nuclear fuels range from a few years to more than 10,000 years. This means that while the radioactivity of nuclear waste initially decreases rapidly, some radioactivity remains for a very long time. Outline the storage of spent nuclear fuels in both the short and long term. *[2]*

Short term:

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Long term:

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(Option C continues on the following page)

(Option C continued)

12. Energy production presents many threats to the environment. One issue that has caused much controversy over recent years is the emission of greenhouse gases, which most scientists believe is a major cause of global warming.

(a) Explain how greenhouse gases affect the temperature of the Earth's surface. *[3]*

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(b) Explain the molecular changes that must occur in order for a molecule to absorb infrared light. *[2]*

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(Option C continues on the following page)

(Option C, question 12 continued)

(c) (i) Carbon dioxide and water vapour are the most abundant greenhouse gases. Identify **one** other greenhouse gas and a natural source of this compound. *[1]*

Greenhouse gas:

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Natural source:

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(ii) Even though water vapour is the more potent greenhouse gas, there is greater concern about the impact of carbon dioxide. Suggest why this is the case. *[1]*

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End of Option C

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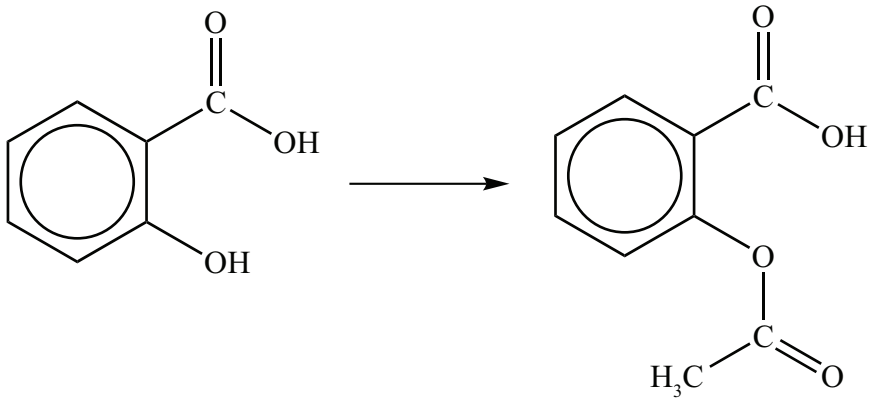
will not be marked.

Option D — Medicinal chemistry

13. Salicylic acid has been used to relieve pain and reduce fevers for centuries, although it can be irritating to the stomach. In the 1800s it was discovered that converting it into acetylsalicylic acid reduces the stomach irritation while still allowing it to be effective.



Salicylic acid Acetylsalicylic acid (aspirin)

(a) Identify the type of reaction used to convert salicylic acid to acetylsalicylic acid. *[1]*

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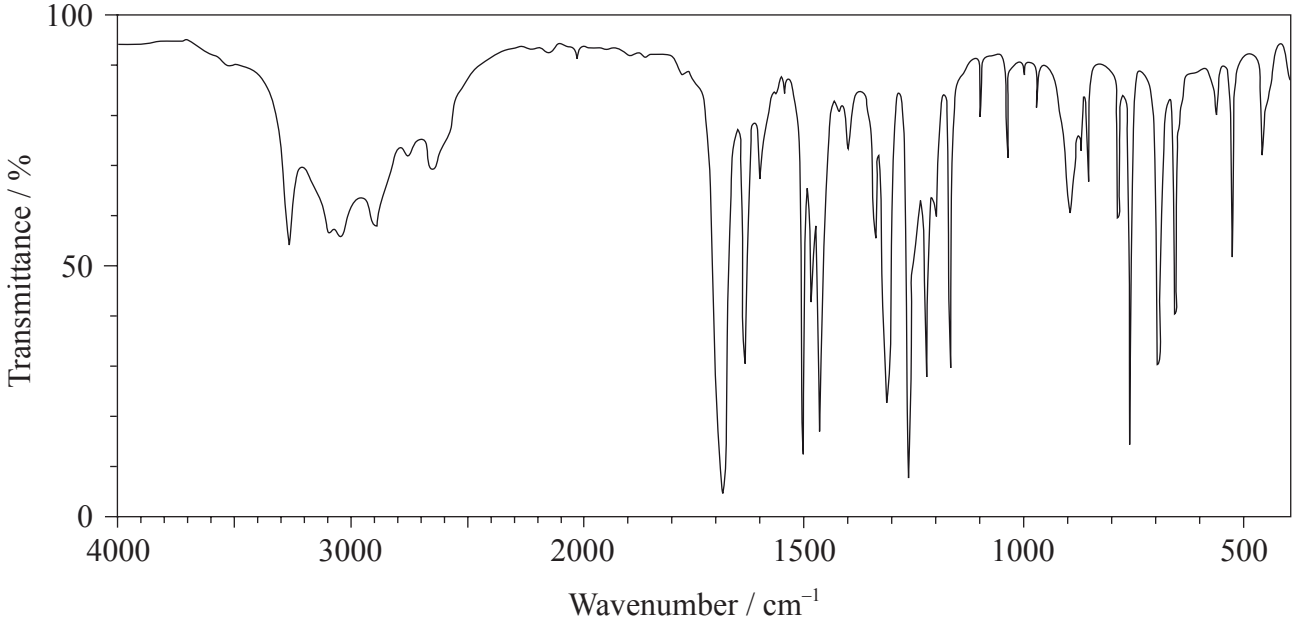
(Option D continues on the following page)

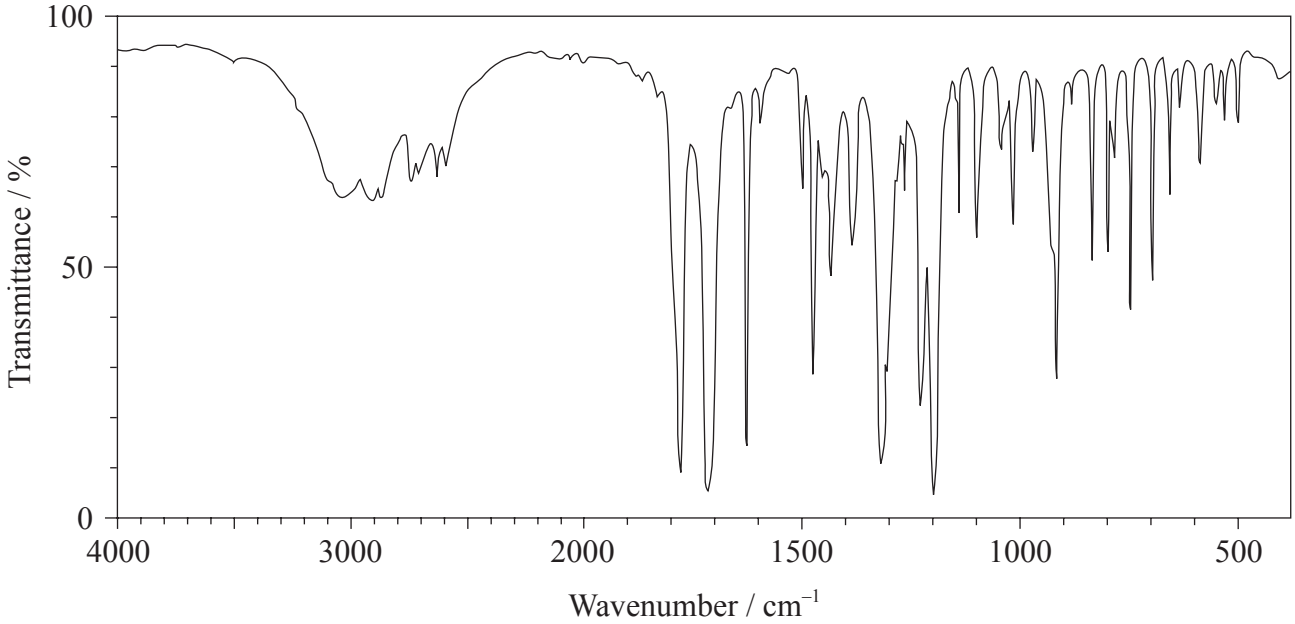
(Option D, question 13 continued)

(b) The infrared (IR) spectra for salicylic acid and acetylsalicylic acid are shown below.

Salicylic acid



Acetylsalicylic acid



[Source: SDBS web: www.sdbs.riodb.aist.go.jp (National Institute of Advanced Industrial Science and Technology, 2014)]

(Option D continues on the following page)

(Option D, question 13 continued)

Using information from section 26 of the data booklet, compare and contrast the two spectra with respect to the bonds present. *[3]*

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(c) A modified version of aspirin is sometimes made by reacting it with a strong base, such as sodium hydroxide. Explain why this process can increase the bioavailability of the drug. *[3]*

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(Option D continues on the following page)

(Option D continued)

14. Recent advances in research into the viruses that cause flu have led to the production of two antiviral drugs, oseltamivir (Tamiflu®) and zanamivir (Relenza®).

(a) Outline why viruses are generally more difficult to target with drugs than bacteria. *[1]*

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(b) By reference to their molecular structures given in section 37 of the data booklet, state the formulas of **three** functional groups that are present in both oseltamivir and zanamivir and the formulas of **two** functional groups that are present in zanamivir only. *[3]*

Present in both:

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Present in zanamivir only:

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(c) Comment on how the widespread use of these drugs may lead to the spread of drug-resistant viruses. *[2]*

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(Option D continues on the following page)

(Option D, question 14 continued)

(d) Outline the general processes that should be followed to promote "green chemistry" in the manufacture of drugs such as oseltamivir and zanamivir. *[3]*

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(Option D continues on the following page)

(Option D continued)

15. Antacids help to neutralize excess hydrochloric acid produced by the stomach. The neutralizing power of an antacid can be defined as the amount in moles of hydrochloric acid that can be neutralized per gram of antacid.

(a) Formulate an equation to show the action of the antacid magnesium hydroxide. *[1]*

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(b) An antacid tablet with a mass of 0.200 g was added to 25.00 cm^3 of 0.125 mol dm^{-3} hydrochloric acid. After the reaction was complete, the excess acid required 5.00 cm^3 of 0.200 mol dm^{-3} sodium hydroxide to be neutralized. Determine the neutralizing power of the tablet. *[3]*

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End of Option D